Semester IV **B. Tech CSE (AI)**

BCSAI 406: DATABASE MANAGEMENT SYSTEMS WITH SQL LAB

0L + 0T + 4P + 2C **MM 100**

List of Experiments:

1. Perform following actions using SQL statements
    a. Create a new user with name "shiva" and password "kumar@1"
    b. Assign the following privileges
        i. Create and drop tables
        ii. Create and drop users
        iii. Allow to assign above privileges to new users
        iv. List all tables in the database
        v. List all users in the database
        vi. Logout from current user and log in as "shiva"
2. Create following tables and insert minimum 10 rows in to each table
    a. Department table with following columns with appropriate data types
        i. DeptId
        ii. DeptName
        iii. DeptLoc
    b. Employee table with following columns with appropriate data types
        i. EmpId
        ii. EmpName
        iii. DOB
        iv. DOJ
        v. Job
        vi. Salary
    c. Product table with following columns with appropriate data types
        i. ProdId
        ii. ProdName
        iii. Price
    d. Sales table with following columns with appropriate data types
        i. SalesId
        ii. Date
        iii. Quantity
3. Update above tables with following features using SQL statements
    a. Make DeptId in Department table as Primary Key
    b. Make EmpId in Employee table as Primary Key
    c. Add DeptId column to the Employee table and make it foreign key from Department table and update the values
    d. Add EmpId and ProdId to the Sales table and make them foreign key from Employee and Product table and update the values
    e. Update all columns in all tables with appropriate constraint such as not null, check and so on

4. Perform the following SQL statements
    a. Create a view "EmpDeptView" from Employee and Department table which contains following columns
        i. EmpName
        ii. DOB
        iii. Salary
        iv. DeptId
        v. DeptName
        vi. Loc
    b. Retrieve all employees whose salary between 25,000 to 30,000
    c. Retrieve all employees who is working in Accounts department (If it is not there add this row to Department table)
    d. Retrieve all employees who is working other than Accounts department
    e. Retrieve all employee who is working in Sales department and Bangalore location
    f. Retrieve all employees who completed minimum 5 years
    g. Retrieve all employees who completed minimum 5 years and salary less than 30,000
5. Perform the following SQL statements
    a. Retrieve all employees whose salary more than 30,000
    b. Retrieve employee details who is getting maximum salary
    c. Retrieve employee details who is getting minimum salary
    d. Retrieve employee details who is getting 3rd maximum salary
    e. Retrieve employee details who is getting 5th minimum salary
    f. Retrieve total number of employees in each department in Bangalore location
    g. Retrieve total number of employees in each location
    h. Retrieve total number of employees in each location in Accounts department
    i. Retrieve total number of employees who complete more than 10 years in each department
6. Write a PL/SQL Procedure to find prime number from 1 to n, n is a user input or parameter
7. Write a PL/SQL Functions to return number of days an employee working using EmpId
8. Write a PL/SQL Procedure to find sum of salaries of all employee working in a particular location
9. Write a PL/SQL Function to return sum of sales by ProdId
10. Write a PL/SQL Function to return sum of sales by EmpId
11. Write a PL/SQL Procedure to generate Employee Report department wise as follows

    DeptName

    EmpName Job Location Salary Cumulative_Salary

12. Write a PL/SQL Trigger to insert row into OldEmployee table when a employee deleted from Employee table (Create OldEmployee table)
13. Write a PL/SQL Trigger not to delete more than 2 employees at a time
14. Write a PL/SQL Trigger not to update employee salary if it cross 67000
15. Write a PL/SQL Package with following procedures and functions
    a. Procedures
        i. Print Total Quantity Sales Summary Report(SalesId, Date, Quantity and Total Quantity)
        ii. Print Total Quantity Sales Summary Report by Date wise
    b. Functions
        i. Return employee name who made maximum sales till date
        ii. Return product name soled maximum quantity till date